C.V. RAMAN COLLEGE OF ENGINEERING, BHUBANESWAR

SUBJECT: MATHEMATICS-II (BS1104)

QUESTION BANK

Unit – III [Ordinary Differential Equations Of First Order] (08 Hours)

First order differential equations, Separable equation, exact differential equation, linear differential equation, Bernoulli's equation and application to Electrical circuits and Orthogonal Trajectories. System of differential equations.

1.	Solve the following equations. (i) $x^2ydy = (x^3 + y^3)\,dy$ (ii) $\left(y + \sqrt{x^2+y^2}\right)dx - xdy = 0, y(1) = 0$ (iii) $(x + y - 3)dy = (x - y + 1)dx = 0$ (iv) $(2x + y + 3)dy = (x + 2y + 3)dx$ (v) $(2x - 4y + 5)dy + (x - 2y + 3)dx = 0$ (vi) $(2x + 3y + 1)dx + (1 - 4x - 6y)dy = 0, y(-2) = -2$
2.	Solve the following equations. (i) $(y\cos x + \sin y + y)dx + (\sin x + x\cos y + x)dy = 0$ (ii) $\left(\frac{3-y}{x^2}\right)dx + \left(\frac{y^2-2x}{xy^2}\right)dy = 0, y(-1) = 2.$ (iii) $(x^2y - 2xy^2)dx + (3x^2y - x^3)dy = 0$ (iv) $(3x^2 - y^2)dx - 2xydy = 0$ (v) $(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)xdy = 0$ (vi) $y(xy + x^2y^2)dx + x(xy - x^2y^2)dy = 0$ (vii) $2\sin(y2)dx + xy\cos(y2)dy = 0,\ y(2) = \sqrt{\frac{\pi}{2}}$ (viii) $2\cosh x\cos y dx = \sinh x\sin y dy$ (ix) $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$ (x) $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$
3.	Solve the following equations. (i) $x\frac{dy}{dx} - (x+1)y = x^2 - x^3$ (ii) $\frac{dy}{dx} - \left(\frac{3}{x} + 1\right)y = x + 2, y(1) = e - 1$ (iii) $(x+1)\frac{dy}{dx} + (x+2)y = 2xe^{-x}$ (iv) $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x}\ln x$ (v) $\cos x dy = y(\sin x - y)dx$ (vi) $(xy^5 + y)dx - dy = 0.$
4.	The rate of decay of a radioactive substance is proportional to the amount present. The half –life period of the sample of the substance is 10 years. How long does it take for 90% of the original amount of substance to disintegrate ?
5.	At earth's surface the velocity of escape is 11.2km/sec. If the projectile is carried by a rocket and is separated from it at a distance of 1000 km from the earth's surface, what would be the minimum velocity at this point sufficient for escape from the earth?
6.	A thermometer reading 5^0 C is brought into a room whose temperature is 22^0 C. One minute later the thermometer reading is 12^0 C. How long does it take until the reading is practically 22^0 C.
7.	A tank contains 400 gal of brine in which 200ln of salt are dissolved. Fresh water runs into the tank at the rate of 2gal/min, and the mixture, kept practically uniform by stirring, runs out at the same rate. How much salt will there be in the tank at the end of 1 hour?
8.	Find the current in the RL circuit with the voltage $E(t) = E_0 \sin \omega t$.
9.	Find the current at any time t > 0 in a circuit having in series a constant electromotive force 40V, a resistor 10Ω, and an inductor 0.2H given that the initial current is zero. Find the current when $E(t) = 150\cos 200t$.
10.	A capacitor c = 0.01F in series with a resistor R = 20 Ohms is charged from a battery E = 10V. Assuming that initially the capacitor is completely uncharged, determine the charge Q(t), voltage v(t) on the capacitor and the current I(t) in the circuit.
11.	A capacitor C = 0.2 farad in series with a resistor R = 200 Ohms is charged from a source E = 24 volts. Find the voltage

	V(t) on the capacitor, assuming that at t = 0, the capacitor is completely uncharged.
12.	Find the orthogonal trajectories of the following families of curves. (a) $y = cx^2$ (b) $x^2 + 4y^2 = c$ (c) $y = c/x^2$
13.	Show tha family of curves $x^2 + 4y^2 = c_1$ and $y = c_2x^4$ are mutually orthogonal.
14.	Find particular member of orthogonal trajectories of $x^2 + cy^2 = 1$ passing through (2,1).
15.	Show that the family of parabolas $y^2 = 4cx + 4c^2$ is self orthogonal.
16.	What is the degree and order of the differential equation whose general solution is $\sin y = ax^2 + be^x$, where a and b are two arbitrary constants.
17.	The amount of bacteria in a test tube doubles in 8 hours. In how many hours will it triple? Assume that the rate of growth is proportional to the amount of bacteria at any time.
18.	Find a real general solution of the following systems of equations. (i) $\frac{dy_1}{dx} = 2y_1 + 2y_2;\ \frac{dy_2}{dx} = 5y_1 - y_2$ (ii) $\frac{dy_1}{dx} = -8y_1 - 2y_2;\ \frac{dy_2}{dx} = 2y_1 - 4y_2$ (iii) $\frac{dy_1}{dx} = -3y_1 - y_2 + 2y_3;\ \frac{dy_2}{dx} = -4y_2 + 2y_3;\ \frac{dy_3}{dx} = y_2 - 5y_3$ (iv) $\frac{dy_1}{dx} = -y_1 - 4y_2 + 2y_3;\ \frac{dy_2}{dx} = 2y_1 + 5y_2 - y_3;\ \frac{dy_3}{dx} = 2y_1 + 2y_2 + 2y_3$.
19.	Solve the following initial value problems involving systems. (i) $\frac{dy_1}{dx} = 2y_1 + 2y_2;\ \frac{dy_2}{dx} = 5y_1 - y_2;\ y_1(0) = 0;\ y_2(0) = -7$ (ii) $\frac{dy_1}{dx} = 2y_1 + 5y_2;\ \frac{dy_2}{dx} = -\frac{1}{2}y_1 - \frac{3}{2}y_2;\ y_1(0) = 10;\ y_2(0) = -5$ (iii) $\frac{dy_1}{dx} = -14y_1 + 104y_2;\ \frac{dy_2}{dx} = -5y_1 + y_2;\ y_1(0) = -1;\ y_2(0) = 1$
	SHORT QUESTIONS
1.	Find the order and degree of the differential equation $y = x\frac{dy}{dx} + \frac{x}{\left(\frac{dy}{dx}\right)}$.
2.	Find the solution of $(y+1)dx - (x+1)dy = 0$.
3.	Find the integrating factor of the differential equation $\left(xy^2 - e^{1/x^3}\right)dx - x^2y\,dy = 0$.
4.	Write the exact differential equation $du = 0$ where $u(x,y) = e^{x^2/y}$.
5.	Write the condition for exactness of of the differential equation $M(x,y)dx = -N(x,y)dy$, where $M(x,y)$ and $N(x,y)$ are arbitrary functions of x and y.
6.	Find the general solution of $y' + y\sin x = xe^{\cos x}$.
7.	What do you mean by integrating factor ? How it helps to solve a differential equation ?
8.	Form a differential equation for a circuit with emf $E = 12V$, resistance $R = 5\Omega$ and $L = 0.01H$.
9.	Find the order and degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{5/2} = \frac{d^3y}{dx^3}$.
10.	Examine whether the differential equation $(x^3 - 3xy^2)dx + (y^3 - 3x^2y) = 0$ is exact.
11.	Write down the differential equation which governs the current in the circuit with $R = 10\Omega$, $L = 0.1H$ and emf $E = 10\sin 50t$.
12.	Is the differential equation $ydx - xdy = 0$ exact? If not, find an integrating factor.
13.	What is Bernoulli's equation? How it can be reduced to a linear equation?
14.	Solve $\frac{dy}{dx} = e^{2x+3y+5}$.
15.	Find the differential equation which has $xy = ae^x + be^{-x} + x^2$ as its solution.
16.	Solve $\tan y \sec^2 x\, dx + \tan x \sec^2 y\, dy = 0$.
17.	Write down the differential equation of the family of circles passing through the origin and having their centers on the x-axis.
18.	What is an orthogonal trajectory?

19.	Find the orthogonal trajectories of $y = kx^2$.
20.	Solve: $x\frac{dy}{dx} = x + y$
21.	Solve: $\frac{dy}{dx} = (x + y)^2$
22.	Solve: $y\frac{dy}{dx} + x = 0; y(1) = \sqrt{3}$
23.	Solve: $\frac{dy}{dx} + cosecy = 0$
24.	Solve: $L\frac{dI}{dt} + RI = 0, I(0) = I_0$